AC Power-Flow Sensitivity-Based Droop Design with Independently Selectable Security Constraints

Methodological working paper | 11 September 2026
Steady-state formulation and proposed validation protocol

Abstract

Volt-var droop settings determine both reactive support and the local response of network voltages to changing injections. This paper develops a steady-state design formulation that incorporates droop feedback directly into the nonlinear AC power-flow Jacobian. The resulting sensitivities describe voltage and reactive-power responses to explicitly scaled active and reactive disturbances. A common physical model supports sensitivity-only design, security-only design, and joint design with independently selectable objectives and sensitivity bounds. Controller parameters are shared across operating cases, while balancing, dispatch recourse and equipment positions are specified separately. The formulation accounts for locally active deadband and saturation regions and distinguishes local derivatives from finite-event security. A reproducible validation protocol is proposed using independently solved AC equilibria, perturbation tests and held-out contingencies. This working paper presents the methodology; numerical performance has not yet been evaluated.

Keywords: volt-var control; droop design; AC power flow; voltage sensitivity; security-constrained optimization; reactive power.

1. Introduction

A droop curve maps a voltage measurement to a reactive-power response. Its slope therefore affects more than the nominal operating point: it also changes how the network responds to disturbances. A design that performs well at one equilibrium may provide weak disturbance rejection, reach reactive limits under a finite event, or fail a contingency constraint. These questions should be examined using a consistent physical model while retaining their distinct meanings.

Network-aware droop synthesis has been studied using linear approximations of AC power flow and robust treatment of uncertain injections [1]. Classical reduced-Jacobian analysis provides a foundation for examining voltage-reactive-power relationships [3]. More recent optimal-rule formulations address bounded curves, deadbands and saturation [4]. Transmission-connected photovoltaic control also provides a relevant sensitivity-based adaptive-deadband precedent [5]. These antecedents motivate a formulation that separates the local response criterion from the finite operating cases used to assess security.

The methodological focus is a common comparison framework rather than a claim that Jacobian reduction is new. The paper derives the droop-modified AC response, defines selectable performance quantities, and formulates matched standalone and joint design problems. It then specifies how the predicted responses and extracted settings should be verified. The analysis concerns steady-state equilibria; it does not establish excitation-system damping or transient stability.

2. AC network and controller model

2.1. Power balance and response policy

Let the network admittance matrix for operating case c be $Y^c = G^c + jB^c$. It includes the declared topology, transformer ratios and phase shifts, and bus shunts. All quantities use a consistent per-unit base. With the case index suppressed, define $\delta_{ij} = \delta_i - \delta_j$. Net injections are positive into the network:

$$P_i(V, \delta) = \sum_j V_i V_j [G_{ij} \cos \delta_{ij} + B_{ij} \sin \delta_{ij}] \tag{1}$$

$$Q_i(V, \delta) = \sum_j V_i V_j [G_{ij} \sin \delta_{ij} - B_{ij} \cos \delta_{ij}] \tag{2}$$

$$P_i = p_i^0 + u_{P,i}, \quad Q_i = q_i^0 + \sum_g C_{ig} q_g + u_{Q,i} \tag{3}$$

The incidence matrix C assigns droop-controlled reactive injections to buses. The fixed term q^0 excludes these injections and excludes shunts already represented in Y. A positive disturbance u increases net injection; an increase in load has the opposite sign.

For the initial derivation, the system is connected and has one slack bus with fixed angle and voltage magnitude. That bus supplies the active and reactive mismatch. At all other buses, active targets are fixed during a local disturbance and voltage magnitudes are unknown. Ideal PV buses, when present, require removal of their fixed-voltage variables and corresponding reactive-balance equations, followed by recovery of reactive outputs. Distributed balancing instead requires explicit participation equations and a balancing variable. These alternatives define different response policies and must not be mixed within a comparison.

2.2. Local and remote voltage droop

Let M select terminal or remote regulated voltage magnitudes. In the unsaturated, zero-deadband region, the controller law is

$$v_g^{reg} = (MV)_g, \quad q_g = q_g^{ref} - k_g[(MV)_g - v_g^{ref}], \quad k_g \geq 0. \tag{4}$$

The coefficient k is the magnitude of dQ/dV, not its reciprocal voltage-droop convention. Fixed-reference voltage contributions are retained as constants when the model is reduced. An ideal fixed-voltage constraint must not also be imposed on the same controlled degree of freedom. A bounded physical curve is represented by

$$q_g = \mathrm{clip}\{q_g^{ref} - k_g d_g((MV)_g - v_g^{ref}), q_g^{min}, q_g^{max}\}. \tag{5}$$

Within a differentiable region, let a_g equal one in an active proportional segment and zero in a constant deadband or constant saturation segment. Define $K_a = \mathrm{diag}(a_g k_g)$. Asymmetric curves use the slope of the active segment. Locally constant capability is assumed here. Voltage-dependent limits, reactive-current compensation and point-of-interconnection flow control require derivatives of their actual equations.

3. Droop-modified power-flow sensitivity

3.1. AC Jacobian

Define the reduced Jacobian blocks $H = \partial P/\partial\delta$, $N = \partial P/\partial V$, $L = \partial Q/\partial\delta$ and $T = \partial Q/\partial V$ after applying the reference and bus-type reductions. Differentiation of (1)-(2) gives, for $i \neq j$,

$$\begin{aligned} H_{ij} &= V_i V_j (G_{ij} \sin\delta_{ij} - B_{ij} \cos\delta_{ij}) \\ N_{ij} &= V_i (G_{ij} \cos\delta_{ij} + B_{ij} \sin\delta_{ij}) \\ L_{ij} &= -V_i V_j (G_{ij} \cos\delta_{ij} + B_{ij} \sin\delta_{ij}) \\ T_{ij} &= V_i (G_{ij} \sin\delta_{ij} - B_{ij} \cos\delta_{ij}). \end{aligned} \tag{6}$$

$$\begin{aligned} H_{ii} &= -Q_i - B_{ii} V_i^2, \quad N_{ii} = P_i/V_i + G_{ii} V_i \\ L_{ii} &= P_i - G_{ii} V_i^2, \quad T_{ii} = Q_i/V_i - B_{ii} V_i. \end{aligned} \tag{7}$$

Here P_i and Q_i are the network injections from (1)-(2). Matrix derivative implementations can follow the sparse AC conventions in [2], with residual signs checked explicitly.

3.2. Incorporating reactive feedback

For fixed settings, fixed equipment and a fixed active region, $dq = -K_a M\, dV$. Differentiating the balances therefore yields the coupled equations

$$\begin{aligned} H\, d\delta + N\, dV &= du_P \\ L\, d\delta + (T + CK_aM)\, dV &= du_Q. \end{aligned} \tag{8}$$

Let A denote the block coefficient matrix of (8), with first block row $[H, N]$ and second block row $[L, T + CK_aM]$. Droop thus changes the reactive-voltage block. Every block is evaluated at the equilibrium associated with the candidate settings; freezing the nominal Jacobian defines a different approximation.

$$\begin{aligned} R = T - LH^{-1}N, \quad R_d &= R + CK_aM \\ R_d\, dV &= du_Q - LH^{-1} du_P. \end{aligned} \tag{9}$$

$$\begin{aligned} S_{VQ} = R_d^{-1}, \quad S_{VP} &= -R_d^{-1} LH^{-1} \\ dq &= -K_a M (S_{VP} du_P + S_{VQ} du_Q). \end{aligned} \tag{10}$$

Equations (9)-(10) require the relevant inverses to exist. In computation, sparse linear solves with A are preferable to explicitly forming inverses. A singular H prevents this Schur reduction even when A is regular. Remote sensing can make the feedback contribution nonsymmetric, so increasing gains need not improve every response measure. Regularity of A is a local steady-state condition, not a dynamic-stability certificate.

4. Disturbance models and design quantities

4.1. Scaled response maps

Let z collect normalized disturbance coordinates and write the stacked injection disturbance as u = Dz. Columns of D specify affected buses, active/reactive directions and physical magnitudes; a column may represent a load change at fixed power factor. Let E_V extract the voltage portion of the state and O select monitored buses. The sensitivity map is computed from

$$AZ = D, \quad S = OE_V Z, \quad dV_{mon} = S\, dz. \tag{11}$$

This definition makes the disturbance assumptions part of the study data. Comparing unscaled matrices across different injection bases or bus selections is not meaningful. Correlation may be represented by the covariance of z, while D specifies the physical disturbance directions.

4.2. Mean-square and worst-case objectives

For zero-mean disturbances with identity covariance, a weighted mean-square voltage-response objective is the squared Frobenius norm. A general covariance Σ gives

$$\begin{aligned} J_s &= \|W_V S\|_F^2 \\ J_\Sigma &= \mathrm{tr}(W_V S \Sigma S^T W_V{}^T). \end{aligned} \tag{12}$$

For bounded disturbances, the appropriate induced norm depends on the uncertainty set. Euclidean disturbances and componentwise box disturbances yield, respectively,

$$\begin{aligned} \max\{\|W_V S z\|_2 : \|z\|_2 \le 1\} &= \|W_V S\|_2 \\ \max\{\|W_V S z\|_\infty : \|z\|_\infty \le 1\} &= \max_i \textstyle\sum_j |(W_V S)_{ij}|. \end{aligned} \tag{13}$$

Direction-specific limits can instead bound individual entries of S. The largest matrix entry does not represent the worst simultaneous box disturbance. The choice of norm, voltage weights, disturbance magnitudes and aggregation across operating cases is fixed before comparing designs.

4.3. Reactive effort and parameter robustness

Voltage response alone may favor aggressive gains, increase reactive effort or reduce headroom. The reactive-response matrix S_q should therefore be evaluated alongside voltage sensitivity, capability use and saturation incidence:

$$S_q = -K_a M E_V Z. \tag{14}$$

Sensitivity to installation errors in the slopes is a separate quantity. For the unsaturated zero-deadband law, let X_k be the derivative of the stacked state with respect to k. Its active-balance right-hand side is zero and its reactive-balance right-hand side is

$$(AX_k)_P = 0, \quad (AX_k)_Q = -C\,\mathrm{diag}(MV - v^{\mathrm{ref}}). \tag{15}$$

Equation (15) must not replace disturbance sensitivity. It answers a different robustness question and must be rederived for other curve regions or adjustable references.

5. Standalone and joint design formulations

Let k be shared across a set of design cases C. Each case has a separate equilibrium, availability pattern and declared recourse r_c. Bounds on k define the installable design family. A nominal-deviation objective can be written as

$$J_{dev} = \sum_g w_g [(k_g - k_g^{nom})/k_g^{scale}]^2. \tag{16}$$

Alternatively, J_0 may represent active losses or another explicitly defined performance quantity. The nominal references and normalization scales are study inputs, separate from numerical initial guesses. Every design enforces its selected AC equations and controller bounds. Security cases additionally enforce the declared operating limits, including

$$\begin{aligned} V_i^{min} \le V_{i,c} \le V_i^{max} \\ P_{ij,c}^2 + Q_{ij,c}^2 \le (S_{ij}^{max})^2. \end{aligned} \tag{17}$$

Branch limits apply at both ends where required; generator limits, outages and corrective-action bounds are also explicit. A contingency does not authorize independent retuning of installed slopes. Table 1 defines the principal comparisons.

Table 1. Design variants using common physical and evaluation models.

Variant	Design cases	Criterion
Nominal	Evaluation only	Prescribed settings
Sensitivity-only	Selected base states	Minimize J_s with base physics and limits
Security-only	Base and contingencies	Minimize J_0 with security limits
Joint objective	Base and contingencies	Minimize J_s with security limits
Joint bounds	Base and contingencies	Minimize J_0; enforce sensitivity thresholds

5.1. Sensitivity as a hard requirement

For the joint bounded variant, choose a sensitivity case subset C_s and impose the following constraints in addition to AC balances, control laws, equipment bounds and security limits:

$$\begin{aligned} A_c Z_c = D_c, \quad S_c = OE_V Z_c \\ J_{s,c} \le \varepsilon_c, \quad \text{for each } c \text{ in } C_s. \end{aligned} \tag{18}$$

The optimization minimizes J_0 over common slopes, case states, permitted recourse and auxiliary sensitivities. An epsilon sweep exposes the tradeoff between sensitivity and the primary cost. Sensitivity-only design omits contingency selection from its training criterion but retains base-case physics and limits. A sensitivity initializer or weighted penalty alone does not establish final compliance with (18).

The auxiliary formulation avoids explicit matrix inverses. Since A depends on the state and slopes, differentiating (18) requires second derivatives of the original equilibrium equations. An outer method using factorization-based sensitivity evaluation is another implementation option. Both approaches remain nonconvex; a local solution is not a global optimality certificate.

6. Numerical treatment and independent verification

6.1. Active regions and equipment response

A derivative is valid locally within its controller active region. At deadband or saturation breakpoints, one-sided responses and finite-perturbation tests must identify the entered branch. Sensitivities of a smoothed numerical encoding are recorded separately from those of the exact physical curve. Voltage- or current-dependent capability contributes additional tangent terms and cannot be represented by zero response solely because a limit is active.

Tap and switched-shunt positions remain fixed during the initial local disturbance analysis, even when their nominal positions were selected by OPF. Subsequent switching is evaluated through a separately specified sequential policy. Differentiating a continuous equipment relaxation does not describe physical discrete actions. Dispatch targets are similarly fixed except for the declared balancing response.

6.2. Computational procedure

Step	Operation and recorded evidence
1	Solve candidate AC equilibria using shared controller settings and declared recourse.
2	Identify exact active regions; assemble A and D with the correct reference and balancing policy.
3	Solve AZ = D; record linear-system residuals, scaled conditioning and invalid cases.
4	Optimize selected objectives and bounds; extract ordinary physical controller settings.
5	Freeze settings, re-solve the AC cases and independently evaluate physics and design criteria.
6	Test positive/negative perturbations, active-region transitions and held-out contingencies.

Near singularity, report conditioning and response amplification rather than introduce an undisclosed regularizer. Failure of a local nonlinear solver does not prove infeasibility. Inconsistent or unavailable derivatives remain explicit missing results rather than being replaced by zero. Operational soft-limit studies must also be distinguished from secure solutions; artificial nodal residuals cannot establish physical feasibility.

6.3. Perturbation validation

For disturbance direction d, perturb injections by ±hDd and obtain fresh AC solutions with frozen settings. Inside a smooth region, compare the central difference with the predicted response:

$$e(h) = \|O[V(u + hDd) - V(u - hDd)]/(2h) - Sd\|_2. \tag{19}$$

Use several decreasing step sizes and check the expected convergence trend before solver tolerance and roundoff dominate. Absolute error and appropriately scaled relative error should both be reported. At a kink, replace this central-difference interpretation with one-sided tests; a central difference may average distinct branches.

Small fixtures should exercise zero gain, proportional response, deadband, saturation, multiple generators per bus, remote sensing, disturbance signs and balancing changes. Analytic or automatic derivatives should be checked against independent numerical derivatives. End-to-end checks require new power-flow solutions, not only a second solve of the same linear system.

7. Proposed empirical evaluation

7.1. Matched designs and test sets

The initial empirical study should use a small interpretable AC network followed by a transmission benchmark whose material equipment is represented. The comparison should include nominal settings, frozen-Jacobian tuning, nonlinear sensitivity-only design, security-only design and joint design. References, slope bounds, controllable devices, design cases, recourse, solver tolerances and evaluation cases must be matched. Multiple starting points and computational budgets should be reported.

Training and evaluation cases are separated before tuning. Held-out tests freeze installed settings and use the same balancing and corrective-action permissions across methods. To isolate the effect of slope design, the first comparison keeps reference settings fixed. Changes caused by optimized dispatch should be distinguished from changes caused by the slopes themselves.

7.2. Performance measures and ablations

The evaluation should report voltage-response norms, reactive effort, available headroom, saturation incidence, active losses and nominal-setting deviation. Security reporting includes both constraint-level violations and the number of affected scenarios, together with severity and coverage. Failed or unsupported cases are retained in the denominator and classified separately; they cannot be counted as zero violations.

The central ablations are removal of sensitivity criteria, removal of contingency design cases, and replacement of the re-evaluated Jacobian by a frozen nominal approximation. Further studies vary the monitored buses, disturbance directions, weighting, operating conditions and sensitivity thresholds. Fixed versus optimized tap/shunt positions form an additional factor. Mechanical action metrics require an initial position and an explicit trajectory or sequential policy.

The planned evidence comprises prediction-error curves against perturbation magnitude, sensitivity matrices before and after design, security-cost versus sensitivity-threshold curves, and held-out performance tables. Active-region transitions should be shown alongside large prediction errors to reveal when a local approximation ceases to describe the finite response. No numerical findings are reported in this methodological version.

8. Discussion and conclusion

The derivation identifies the feedback term CK_aM as the direct contribution of locally active droop to the reactive-voltage Jacobian block. Its consequences depend on the network, regulated location, operating point and active curve region. This dependence motivates recomputing sensitivities at each candidate equilibrium rather than treating network-only coefficients as invariant design data.

Sensitivity and security remain complementary. Sensitivity describes small disturbances around a regular equilibrium; security tests specified finite operating cases and their constraints. A design can be assessed against either criterion independently or against explicit bounds on both. Neither a low sensitivity objective nor a regular power-flow Jacobian alone demonstrates transient stability, successful automatic switching, or security against untested events.

The formulation can be implemented as an optional analysis and design layer over shared steady-state physics. Additional objectives reuse the same quantity definitions, while dynamic control and automatic equipment policies retain separate evaluators. The next empirical step is validation of the derivative layer followed by the matched standalone and joint studies specified above. Performance and novelty conclusions require those results and a complete review of the closest prior formulations.

References

[1] K. Baker, A. Bernstein, E. Dall'Anese, and C. Zhao, "Network-Cognizant Voltage Droop Control for Distribution Grids," *IEEE Transactions on Power Systems*, vol. 33, no. 2, pp. 2098-2108, 2018. https://doi.org/10.1109/TPWRS.2017.2735379

[2] R. D. Zimmerman, "AC Power Flows, Generalized OPF Costs and their Derivatives using Complex Matrix Notation," *MATPOWER Technical Note 2*, 2010. https://matpower.org/docs/TN2-OPF-Derivatives.pdf

[3] B. Gao, G. K. Morison, and P. Kundur, "Voltage Stability Evaluation Using Modal Analysis," *IEEE Transactions on Power Systems*, vol. 7, no. 4, pp. 1529-1542, 1992. https://doi.org/10.1109/59.207377

[4] I. Murzakhanov, S. Gupta, S. Chatzivasileiadis, and V. Kekatos, "Optimal Design of Volt/VAR Control Rules for Inverter-Interfaced Distributed Energy Resources," arXiv preprint, 2022. https://arxiv.org/abs/2210.12805

[5] W. Kim, S. Song, and G. Jang, "Droop Control Strategy of Utility-Scale Photovoltaic Systems Using Adaptive Dead Band," *Applied Sciences*, vol. 10, no. 22, article 8032, 2020. https://doi.org/10.3390/app10228032

[6] S. Gupta, V. Kekatos, and S. Chatzivasileiadis, "Optimal Design of Volt/VAR Control Rules of Inverters using Deep Learning," arXiv preprint, 2022. https://arxiv.org/abs/2211.09557

[7] J. Wei, S. Gupta, D. C. Aliprantis, and V. Kekatos, "A Chance-Constrained Optimal Design of Volt/VAR Control Rules for Distributed Energy Resources," arXiv preprint, 2023. https://arxiv.org/abs/2306.06467

Related extensions

Learning-based rule synthesis [6] provides a possible alternative design adapter, while chance-constrained rule design [7] motivates later uncertainty-aware comparisons. These works concern inverter/distribution settings and do not directly establish the transmission synchronous-machine formulation or guarantees considered here.

Manuscript status and source review

This document is a methodological working paper prepared from the project research draft. It contains no experimental results and assigns no author or institutional affiliation. The bibliography identifies the primary methodological references; some literature comparisons rely on abstracts and publication records rather than complete derivations. A submission-ready novelty statement requires full-text review, particularly of the reduced-Jacobian and adaptive-deadband studies. The Jacobian identities and proposed comparison framework are presented under the assumptions stated in this paper.